A Coupled Radiative–Microphysical Model of Titan's Organic Haze: Energy Balance, a Fractal-Aggregate Scaling Law, and Their Feedback

The Titan Haze Feedback Project

Working note — compiled June 10, 2026

Abstract

Titan's organic haze is radiatively and microphysically active: it absorbs sunlight and warms the stratosphere, emits in the thermal infrared, and its vertical and size structure is set by coagulation and sedimentation of fractal aggregates whose rates depend on the local temperature. This makes the haze a genuine feedback element rather than a passive tracer. We outline a four-stage program to isolate this feedback in a one-dimensional column: (i) a correlated-k radiative energy balance for the temperature profile (a mature TAM-lineage engine, cross-validated source-by-source against an independent multi-stream DISORT/`pydisort` implementation); (ii) an analytic *scaling law* for the aerosol microphysics that maps temperature, monomer radius, and fractal dimension onto the haze number- and size-profile; (iii) iteration of (i) and (ii) to a self-consistent fixed point to diagnose the feedback; and (iv) coupling of photochemical haze production into the loop. Our central methodological result is the scaling law. Adopting a monodisperse, self-similar closure $C(z,N) = n(z)\,\delta(N - \bar{N}(z))$ for the concentration of aggregates of N monomers at altitude z, the population balance reduces from an integro-differential equation in (z,N) to a compact boundary-value problem in z for the number density $n(z)$ and mean size $\bar{N}(z)$. In the sedimentation-dominated limit it collapses further to a single first-order ordinary differential equation, $\mathrm{d}\bar{N}/\mathrm{d}z = -\beta(\bar{N}, z)\, P/[2\,\omega(\bar{N}, z)^2]$, whose asymptotic solutions grow as $\bar{N} \propto (\mathrm{depth})^2$ in the free-molecular upper haze and $\bar{N} \propto (\mathrm{depth})^{1/2}$ in the continuum lower atmosphere. Solving the full eddy-diffusion problem reproduces the haze extinction scale height of Tomasko et al. (2008) (64 vs. 65 km) and a sub-micron characteristic radius consistent with de Trenquelléon et al. (2025). This law is inexpensive enough to embed inside the radiative iteration, closing the feedback loop. Doing so yields the central result: *the feedback regime of the coupled system is set by the haze infrared opacity.* With laboratory-tholin IR optical constants (Khare et al. 1984), the model sits in a spurious strong-feedback regime—naive coupled iteration oscillates between a warm ($\sim$183 K) and a cool ($\sim$131 K) stratopause, the fixed-haze radiative transfer is genuinely multi-valued (17 K, two independent engines), and the coupled fixed point, found by a damped branch-tracking (continuation) solve of $T = F(\mathrm{haze}(T))$, is 40 K colder than the observed-like prescribed-haze state. The cause is that lab tholin over-absorbs the 300–900 cm^{-1} thermal window by 20–90× relative to the observationally constrained haze: the stratosphere cools an order of magnitude too efficiently per unit shortwave heating. Calibrating the haze LW spectral absorptivity per unit visible extinction to the observational haze (a composition property, taken like $\omega_0(\lambda)$ and $g(\lambda)$ from observations) raises the equilibrium by +53 K and transforms the coupling: the composite map becomes nearly flat (slope ≈ 0.1), every branch-tracked chain—monodisperse or polydisperse, warm- or cool-started— converges to the same $\sim$194 K stratopause (monodisperse warm/cool split 0.1 K), in agreement with the prescribed-haze references (195 K Fortran engine, $\sim$200 K DISORT engine), and no multiplicity or oscillation survives. Titan's haze–climate coupling is benign; treating it with laboratory IR optical constants manufactures a dramatic feedback that is not there.

1 Introduction

On Titan the photochemical haze is not a passive optical load. It absorbs solar radiation—heating the stratosphere—and emits in the thermal infrared, where it contributes a non-negligible fraction of the planetary emission and warms the troposphere (Tomasko et al., 2008). At the same time its vertical extent and particle sizes are governed by aerosol microphysics: monomers produced high in the atmosphere coagulate into fractal aggregates that sediment downward (Cabane et al., 1993; Rannou et al., 2004; Burgalat and Rannou, 2017). Crucially, the microphysical rates are temperature dependent—the Brownian coagulation kernel scales as T/η, the gas viscosity $\eta(T)$ and mean free path $\lambda(T, P)$ enter the settling velocity, and the free-molecular kernel carries a $T^{1/2}$ factor. Temperature sets the microphysics, the microphysics sets the opacity, and the opacity sets the temperature: a closed feedback loop.

Most existing models break this loop in one direction. Radiative studies prescribe the haze from observations (Tomasko et al., 2008; Lombardo and Lora, 2023), while fully coupled treatments embed microphysics in a three-dimensional general circulation model (de Batz de Trenquelléon et al., 2025), which is faithful but expensive and difficult to interrogate. Our aim is intermediate: to isolate the one-dimensional radiative↔microphysical feedback with a fast, analytic microphysics so the coupled system can be iterated cheaply to a self-consistent state and the feedback gain measured directly.

2 Model overview

The program proceeds in four stages.

Step 1 — Radiative energy balance. A 1-D plane-parallel Titan column iterated to radiative(−convective) equilibrium for $T(z)$: shortwave absorption by CH_4 and haze; longwave cooling by gas lines, collision-induced absorption, and haze emission (Sec. 3). The production engine is a mature TAM-lineage correlated-k model (`example_bowen_fort`); we additionally built an independent multi-stream discrete-ordinates implementation (`pydisort`/DISORT; Stamnes et al., 1988) and matched the two source-by-source, so the engine is cross-validated rather than taken on trust (Sec. 3.2).

Step 2 — Microphysical scaling law. A closed relation for the steady aggregate population, coagulation + sedimentation → density/size profile, in the form $C(z, N)\,\mathrm{d}z\,\mathrm{d}N = f(z, N, T, d, D)$ (Sec. 4). This is the methodological core of the present note.

Step 3 — Coupled iteration. Map the microphysical moments to layer optical properties, feed the RT engine, update $T(z)$, and iterate; diagnose the feedback. The feedback regime proves to be set by the haze IR opacity: lab-tholin optical constants produce a spurious strong-feedback, cold-biased regime, while the observationally calibrated haze gives a benign, monostable coupling at a $\sim 194\,\mathrm{K}$ stratopause (Sec. 5).

Step 4 — Photochemistry. Promote the imposed haze production rate to an output of a photochemical module, closing the radiation↔chemistry↔microphysics loop (Sec. 6).

The Step 1 engine is the reference correlated-k model; our DISORT implementation serves as an independent multi-stream cross-check. The two are matched to within a few percent in every opacity source on a shared 100-layer grid (Sec. 3.2), so the multi-stream solver both validates the engine and provides a higher-angular-resolution option for the scattering-dominated shortwave.

3 Radiative energy balance (Step 1)

The column spans the surface to $\sim 540\,\mathrm{km}$. Following the spectral split of Lombardo and Lora (2023), shortwave (2000–$40000\,\mathrm{cm}^{-1}$, $< 5\,\mu\mathrm{m}$) radiative transfer treats solar absorption and multiple scattering by haze (single-scattering albedo ω_0, asymmetry g), while the longwave band (1–$2000\,\mathrm{cm}^{-1}$) handles thermal cooling with gas line opacity, N_2–N_2, N_2–CH_4, CH_4–CH_4, and N_2–H_2 collision-induced absorption, and a (near-)purely-absorbing haze. Gas opacities use the correlated-k method with HITRAN line data; methane shortwave coefficients follow the Huygens/DISR measurements supplemented above 1600 nm by laboratory data. The temperature profile is relaxed until shortwave heating balances longwave cooling.

The validation target is the energy budget of Tomasko et al. (2008): a surface net solar flux of $0.574\,\mathrm{W\,m^{-2}}$ (about 11% of the incident flux) rising to $\sim 4.1\,\mathrm{W\,m^{-2}}$ near 400 km; a net heating that exceeds cooling by at most $\sim 0.5\,\mathrm{K}$ per Titan day near 120 km; and a thermal cooling rate peaking near 21 K per Titan day around 340 km. Roughly 80% of the top-of-atmosphere sunlight is absorbed in the column, $\sim$40% below 80 km, the haze raising the planetary thermal emission by $\sim$15% and reducing the tropospheric cooling rate by $\sim$30% below 50 km. Baseline atmospheric and compositional inputs are collected in Table 2.

3.1 Implementation and first results

The solver is built on `pydisort` (a modern `torch`-based DISORT binding) with eight streams: a collimated solar beam in the shortwave and a band-resolved Planck thermal source in the longwave. Per layer the optical properties combine the haze and the gas. The haze extinction follows from the Step 2 microphysics through *mean-field (Rayleigh–Debye–Gans) aggregate optics*: each monomer absorbs in the Rayleigh regime, so the aggregate absorption is additive over monomers, $C_{\mathrm{abs}} = N\, C_{\mathrm{abs}}^{\mathrm{mono}}(\lambda)$ with $C_{\mathrm{abs}}^{\mathrm{mono}} = (8\pi^2 d^3/\lambda)\,\mathrm{Im}\{(m^2-1)/(m^2+2)\}$ and $m(\lambda)$ the tholin index (Khare et al., 1984; de Batz de Trenquelléon et al., 2025). This scales the extinction with monomer number (i.e. mass), not with r_a^2, and so corrects the mobility-radius cross-section that overestimated the optical depth roughly eightfold; with the monomer absorption anchored to the observed haze opacity, the visible column optical depth is ≈ 8, matching Huygens/DISR. Spectral single-scattering albedo $\omega_0(\lambda)$ and asymmetry $g(\lambda)$ come from the prescribed haze (Doose 2016): $\omega_0 \to 0$ in the UV (pure absorber), ~ 0.9 in the near-IR, near-pure absorber in the thermal IR. In the thermal IR the haze absorptivity is *not* taken from the laboratory index: as §5.4 shows, lab tholin over-absorbs the thermal window by 20–90×; the LW opacity is instead calibrated to the observational haze through its spectral absorptivity per unit visible extinction, $\chi(\tilde{\nu}, P)$.

The longwave combines, on a common 40-band IR grid, the collision-induced continuum (HITRAN coefficients; Karman et al., 2019; the N_2–N_2 roto-translational band is the origin of Titan's pressure-induced greenhouse), correlated-k gas lines for the stratospheric coolants (CH_4/C_2H_2/C_2H_6/C_2H_4/HCN), and the pure-absorber haze; the shortwave treats CH_4 with the same correlated-k tables as the reference Fortran model under the Houghton solar spectrum at 9.5 AU. The layer-mean temperature is relaxed to radiative–convective equilibrium with convective adjustment and a cold-space top boundary. Grids, k-tables, solver relaxation, and boundary details are collected in SI S1.

The solver reaches a genuine steady state with energy closed to ~ 0.1–$1\,\mathrm{W\,m^{-2}}$ (Fig. 1). Driven by the observationally-prescribed haze it reaches a $\sim 200\,\mathrm{K}$ stratopause, matching the reference Fortran model run on the same haze (Fig. 2, green vs. black)—the radiation physics is validated end-to-end. Driven by the Step 2 *microphysics* haze with the calibrated LW opacity (§5.4) it reaches $\sim 194\,\mathrm{K}$, with the right vertical structure throughout (cold tropopause near 74 K, $\sim$92 K surface).

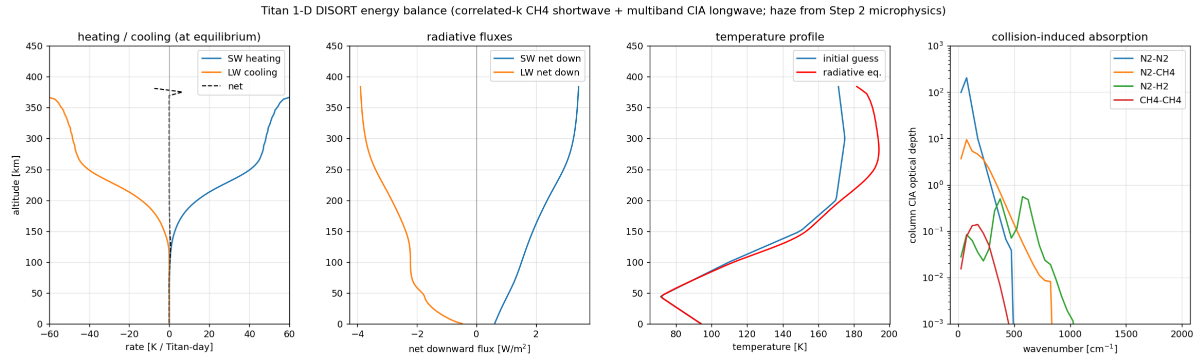

Figure 1: DISORT energy balance on the Titan column with the Step 2 haze and HITRAN collision-induced absorption. From left: shortwave heating, longwave cooling, and net rate at the relaxed state; net downward shortwave and longwave fluxes; initial-guess vs. radiative-equilibrium temperature (surface fixed); and the column CIA optical depth per pair, showing the N_2–N_2 far-IR continuum that dominates the thermal opacity.

3.2 Cross-comparison with a reference radiation model

As an independent check we compare against a mature Fortran radiation model of the TAM lineage (Lombardo and Lora, 2023, the same scheme family as Lombardo & Lora 2023), a full correlated-k treatment with prescribed haze, run to radiative–convective equilibrium on the *same* 100-layer log-pressure grid (1 Pa top) under diurnally-averaged insolation, so the comparison is not confounded by vertical resolution. Figure 2 overlays the equilibria on a shared pressure coordinate: driven by the *prescribed* haze—the observational optical depths the reference uses, in our radiative engine—the DISORT column reaches $\sim$200 K and tracks the reference closely, validating the engine; driven by the Step 2 microphysics haze (calibrated LW, §5.4) it reaches $\sim$194 K, the residual offset reflecting the microphysics haze's remaining visible-opacity excess aloft rather than the radiation.

Beyond the profile overlay, the two engines were matched *source by source*: at the same state, every opacity contribution—the shared prescribed haze (machine-precision agreement), correlated-k CH_4 (column ratios 1.03–1.08), gas lines, Rayleigh, and, after adopting the reference's exponential-sum CIA transmission fits in place of band-averaged cross-sections, the collision-induced continuum—agrees to within a few percent through the column (SI S2). The engines differ only in the radiative-transfer solution itself, not in the optical depths fed to it.

4 Microphysical scaling law (Step 2)

We seek the steady population $C(z, N)$ of aggregates containing N monomers at altitude z (positive upward), built from production, Brownian coagulation, and gravitational sedimentation, with T the temperature, d the monomer radius, and D the fractal dimension. The physical ingredients—kernels, settling velocity, and fractal radius laws—follow Burgalat and Rannou (2017), with parameters from de Batz de Trenquelléon et al. (2025).

4.1 Monodisperse, self-similar closure

We adopt a monodisperse closure: at each altitude the aggregates share a single characteristic monomer count $\bar{N}(z)$,

$$C(z, N) = n(z)\,\delta\big(N - \bar{N}(z)\big)\,, \tag{1}$$

so the unknown fields are the number density $n(z)$ [m^{-3}] and the mean size $\bar{N}(z)$. The count $\bar{N}$ is treated as a *continuous monomer-volume count*: it measures particle volume in units of

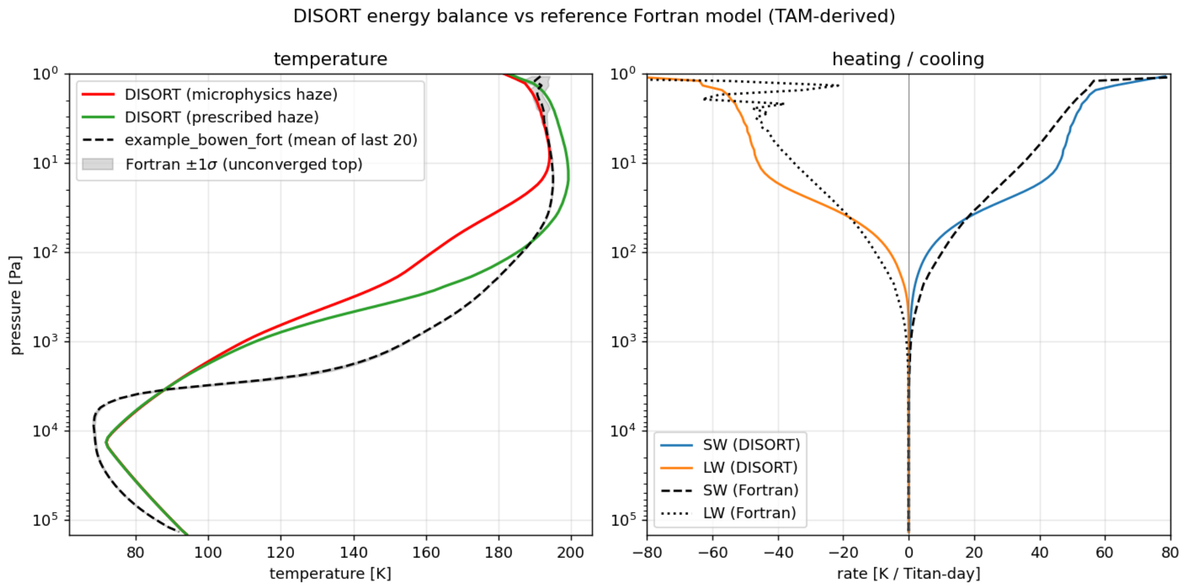

Figure 2: Our DISORT energy balance with the Step 2 *microphysics* haze (red, calibrated LW) and with the observational *prescribed* haze (green), versus the reference TAM-lineage correlated-k Fortran model (black, dashed), on a shared pressure axis. In the stratosphere the prescribed-haze runs of the two engines bracket a $\sim$195–200 K stratopause band (residual solver and boundary differences); the microphysics-haze equilibrium falls at its edge. Right: shortwave heating and longwave cooling. Below $\sim 10^3$ Pa the radiative relaxation time exceeds the integration length for both models, so each retains its own initial state there; the meaningful comparison is the stratosphere. (The reference's explicit stepper formerly oscillated by up to $\sim$24 K near the 1 Pa top; a per-layer step cap reduces this to $\sim$2.5 K.)

one monomer's volume, so $\bar{N} < 1$ denotes a sub-monomer sphere of radius $r = d\,\bar{N}^{1/3} < d$. This keeps mass bookkeeping exact through both coalescence (merging spheres, $\bar{N} < 1$) and aggregation (sticking, $\bar{N} \geq 1$). The closure is exact for the moments $M_0 = n$ and $M_1 = n\bar{N}$; it approximates the kernel by evaluating it at the mean size.

4.2 Fractal geometry and transport coefficients

An aggregate of N monomers of radius d and fractal dimension D has a mass (volume-equivalent) radius and a mobility (drag) radius

$$r(N) = d\,N^{1/3}, \qquad r_a(N) = d\,N^{1/D}, \qquad V(N) = \tfrac{4}{3}\pi d^3 N, \tag{2}$$

with the apparent–bulk conversion $r_a = E^{-1} r^{3/D}$, $E = d^{(D-3)/D}$ (Burgalat and Rannou, 2017). We carry two growth phases explicitly,

$$D(\bar{N}) = \begin{cases} 3 & \bar{N} < 1 \quad \text{(compact coalesced spheres, } E = 1,\ r_a = r\text{)}, \\ D & \bar{N} \geq 1 \quad \text{(fractal aggregates, e.g. } D = 2\text{)}, \end{cases} \tag{3}$$

the switch at $\bar{N} = 1$ being continuous in r_a (both give $r_a = d$).

The Stokes settling velocity with first-order Cunningham slip, generalized to fractal D (Burgalat and Rannou, 2017, Eq. 29), is

$$\omega(N, z) = \frac{2\rho_p\, g(z)\, d^2}{9\,\eta(z)} \left[N^{(D-1)/D} + \frac{A\,\lambda(z)}{d}\, N^{(D-2)/D} \right], \tag{4}$$

with $A = 1.591$, gas viscosity $\eta(z) = \eta(T)$, mean free path $\lambda(z) = \lambda(T, P)$, gravity $g(z)$, and $D = D(\bar{N})$. The equal-size Brownian coagulation kernel is bridged across the Knudsen regimes by the Fuchs harmonic mean of its continuum and free-molecular forms,

$$\beta_{\rm CO}(N, N) = \frac{2k_BT}{3\eta}\,(r_{ai} + r_{aj})\left(\frac{1}{r_{ai}} + \frac{1}{r_{aj}}\right) \cdot [\text{slip}], \tag{5}$$

$$\beta_{\rm FM}(N, N) = \left(\frac{6k_BT}{\rho_p}\right)^{1/2} (r_{ai} + r_{aj})^2 \left(r_i^{-3} + r_j^{-3}\right)^{1/2}, \tag{6}$$

$$\beta(N, N; z) = \frac{\beta_{\rm CO}\,\beta_{\rm FM}}{\beta_{\rm CO} + \beta_{\rm FM}}. \tag{7}$$

Both ω and β are thus explicit functions of $(\bar{N}, z, T, d, D)$; the temperature enters through k_BT/η, $(k_BT/\rho_p)^{1/2}$, $\eta(T)$, and $\lambda(T, P)$.

4.3 Sedimentation-dominated limit: the master scaling ODE

The steady transport of the number density n and the monomer-volume density $M \equiv n\bar{N}$ (mass $= m_1 M$, $m_1 = \rho_p \frac{4}{3}\pi d^3$) is a two-point boundary-value problem—Gaussian production aloft, Smoluchowski number loss $\frac{1}{2}\beta n^2$, settling, and eddy diffusion—whose full formulation, boundary conditions, and solvers are collected in SI S3. Setting $K \to 0$ yields a closed-form law—both a physical limit and the initial guess for the BVP solver. Monomer-flux constancy (coagulation preserves total monomer volume, so the downward monomer flux below the source is the constant production rate $P = \mathcal{Q}_p/m_1$) gives the classical result that the haze mass density equals the mass flux divided by the fall speed,

$$\omega\, n\, \bar{N} = P \implies n(z) = \frac{P}{\omega\, \bar{N}}, \qquad \rho_h(z) = \frac{\mathcal{Q}_p}{\omega(\bar{N}, z)}. \tag{8}$$

Substituting $\omega n = P/\bar{N}$ into the number balance (11) (below the source) collapses the system to a single first-order ODE for the mean size,

$$\boxed{\frac{\mathrm{d}\bar{N}}{\mathrm{d}z} = -\frac{\beta(\bar{N}, z)\, P}{2\,\omega(\bar{N}, z)^2}}, \qquad \bar{N}(z_0) = N_{\rm seed}, \tag{9}$$

integrated downward and switching $D(\bar{N})$ from 3 to D at $\bar{N} = 1$. Because the right-hand side is negative, $\bar{N}$ grows as the particle falls; the number density and sizes then follow from (8) and (2).

4.4 Asymptotic scalings

Holding the background quantities $(T, \eta, \lambda, g, \rho_p, P)$ roughly constant over a scale height, β and ω become pure powers of $\bar{N}$ and (9) integrates to power laws (Table 1). Just below the production region the haze is free-molecular and the mean size grows as $\bar{N} \propto (\text{depth})^2$; deep in the dense lower atmosphere the continuum kernel is size-independent and growth slows to $\bar{N} \propto (\text{depth})^{1/2}$. The full Fuchs kernel (7) interpolates smoothly between these limits, which is why (9)/(10)–(11) are integrated numerically rather than reduced to a single exponent.

4.5 Numerical solution

We integrate the master ODE (9) downward from $z_0 = 415\,\text{km}$ (seed $N_{\rm seed} = (r_p/d)^3 \approx 8\times10^{-3}$) on the crude Titan reference column, with $\eta(T)$ from a Sutherland law and $\lambda(T, P)$ from kinetic theory (Table 2). Figure 3 shows the result. The mean size grows by seven orders of magnitude from the sub-monomer seed to $\bar{N} \sim 7\times10^4$ at the surface; the spherical→fractal switch at $\bar{N} = 1$

Table 1: Asymptotic scaling exponents for equal-size aggregates, with depth $\equiv z_0 - z$. The spherical phase ($\bar{N} < 1$) uses $D = 3$.

Regime	$\beta(\bar{N}) \propto$	$\omega(\bar{N}) \propto$	$\mathrm{d}\bar{N}/\mathrm{d}z \propto -\bar{N}^s$, $s =$	$\bar{N}$(depth), $D = 2$
Continuum (Kn$\ll$1, low alt.)	$\bar{N}^0$	$\bar{N}^{(D-1)/D}$	$-2(D-1)/D$	$\bar{N} \propto \mathrm{depth}^{1/2}$
Free-molecular (Kn$\gg$1, high alt.)	$\bar{N}^{2/D-1/2}$	$\bar{N}^{(D-2)/D}$	$(6-2D)/D - 1/2$	$\bar{N} \propto \mathrm{depth}^2$

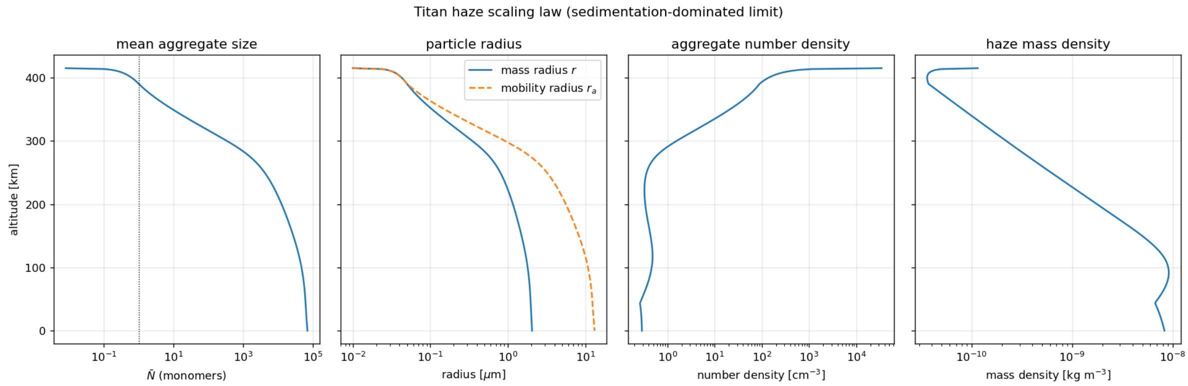

Figure 3: Steady haze profiles from the sedimentation-dominated scaling law (9) on the Titan reference column: mean aggregate size $\bar{N}$ (dotted line marks the $\bar{N} = 1$ spherical/fractal switch), mass and mobility radii, aggregate number density, and haze mass density. Aggregates grow as they settle; r_a exceeds r in the fractal phase.

near 390 km is visible as the divergence of the mobility radius r_a from the mass radius r (open aggregates have $r_a > r$ for $D < 3$). The mass radius reaches $\approx 0.35\,\mu$m near 300 km—consistent with the characteristic radius $r_c \approx 0.46\,\mu$m of de Batz de Trenquelléon et al. (2025)—and $\approx 2\,\mu$m at the surface. Number densities span $\sim$0.3–3 $\times\ 10^4\,\mathrm{cm}^{-3}$, and the monomer mass flux $\rho_h\,\omega$ is conserved to machine precision, confirming Eq. (8).

4.6 Eddy-diffusion BVP and cross-validation

Restoring eddy diffusion, we solve the full boundary-value problem (10)–(11) on $[0, z_0]$, imposing the (narrow) production as a top flux boundary condition—downward monomer flux $\Phi_M = P$ and seed number flux $\Phi_n = P/N_{\mathrm{seed}}$—and settling-only deposition at the surface. The $K \to 0$ master ODE supplies the initial guess. A Titan-like eddy profile $K(z) = K_0 \exp(z/H_K)$ with $K_0 = 1\,\mathrm{m^2s^{-1}}$ and $H_K = 150$ km keeps K small ($\lesssim 16\,\mathrm{m^2s^{-1}}$) across the haze, as expected in Titan's stratosphere.

We cross-validate the resulting profile against two published constraints (Fig. 4). First, the haze extinction scale height: fitting $n\,r_a^2$ over 80–300 km yields $H = 64$ km, against the 65 km inferred by Tomasko et al. (2008); the $K \to 0$ settling limit alone gives 54 km, so the modest eddy diffusion brings the model into close agreement. Second, the characteristic size: the mass radius is $\approx 0.34\,\mu$m near 300 km and sub-micron through the main haze, consistent with the $r_c \approx 0.46\,\mu$m of de Batz de Trenquelléon et al. (2025). The monomer mass flux is conserved to machine precision (global mass balance: surface deposition equals column production), and the BVP departs from the master ODE by only $\sim 9\%$ in the settling-dominated lower haze. The extinction scale height is sensitive to $K(z)$, identifying the eddy profile as the principal uncertain input.

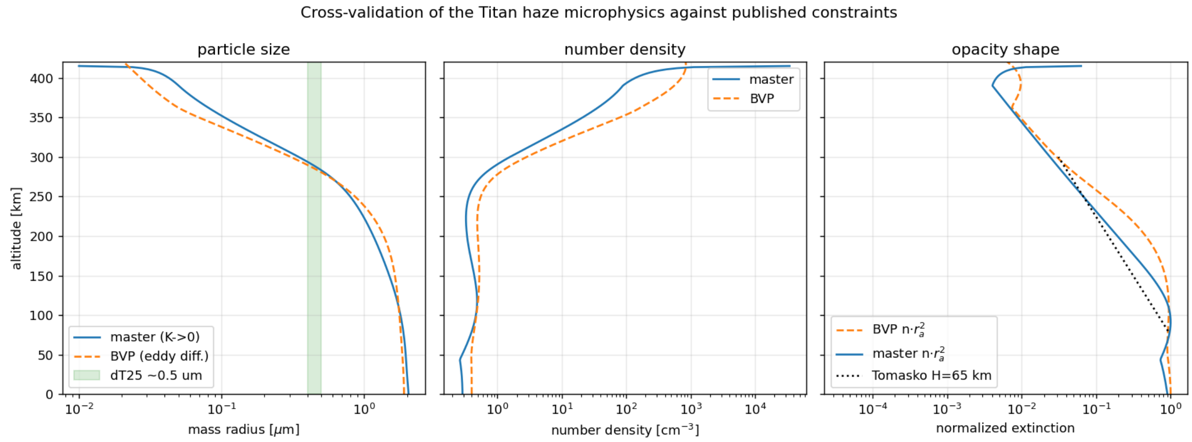

Figure 4: Cross-validation of the microphysics against published Titan constraints. Left: mass radius from the $K \to 0$ master ODE and the eddy-diffusion BVP, with the de Trenquelléon et al. (2025) main-haze size ($\sim 0.5\,\mu$m) shaded. Centre: number density. Right: normalized extinction $n\,r_a^2$, with the Tomasko et al. (2008) $H = 65\,$km reference slope (dotted) anchored at 80 km.

4.7 Polydispersity: a bimodal two-moment extension

The monodisperse closure carries one size per level. To assess what a realistic *size distribution* changes, we extended the scheme to a bimodal two-moment log-normal microphysics following Burgalat and Rannou (2017): a spherical-monomer mode and a fractal-aggregate mode, each carried by its number and volume moments, with Brownian coagulation evaluated by Gauss–Hermite quadrature over the full Fuchs kernel, an inter-mode coalescence transfer, and moment-specific settling velocities (SI S3). The leading new effect is *gravitational sorting*: the moment-averaged settling velocities differ, $\langle\omega\rangle_3 > \langle\omega\rangle_0$, so a broad aggregate distribution sheds its mass faster than a single particle of the mean size and less haze accumulates aloft. The visible column optical depth falls steeply with the aggregate geometric width σ_F (Fig. 5): from $\tau \approx 8.4$ in the monodisperse limit to 6.2 at $\sigma_F = 1.2$ and 1.9 at $\sigma_F = 2.0$. Matching the observed $\tau \approx 8$ requires a narrow distribution, $\sigma_F \lesssim 1.2$, so the haze opacity becomes a *constraint* on the aggregate size spread. The bimodal haze also places its opacity *vertically* much closer to the observational profile than the monodisperse closure (within 0.96–1.3$\times$ below 100 Pa at $\sigma_F = 1.2$, versus the monodisperse 2.6$\times$ excess at 100 Pa).

5 Coupled iteration (Step 3)

Per layer the microphysics supplies the number density $n(z)$ and the characteristic radius $r_a(z) = d\,\bar{N}^{1/D}$. These map to layer optical properties via a precomputed aggregate-optics table (de Batz de Trenquelléon et al., 2025, Eqs. 16–18): the optical thickness is $\Delta\tau(\lambda, z) = n(z)\,C_{\rm ext}(r_a, \lambda)\,\Delta z$, with mean single-scattering albedo and asymmetry parameter obtained by opacity weighting. The aggregate projected area scales as $\bar{N}^{2/D} d^2$. The feedback loop is then

$$T(z) \;\longrightarrow\; \eta(T), \lambda(T, P) \;\xrightarrow{(9)/\S4}\; n(z), \bar{N}(z) \;\xrightarrow{\text{optics}}\; \Delta\tau, \omega_0, g \;\xrightarrow{\text{RT engine}}\; T(z),$$

iterated under-relaxed. The central finding of this section is that the *regime* of this feedback—benign or near-bistable—is set by a single material property, the haze IR opacity.

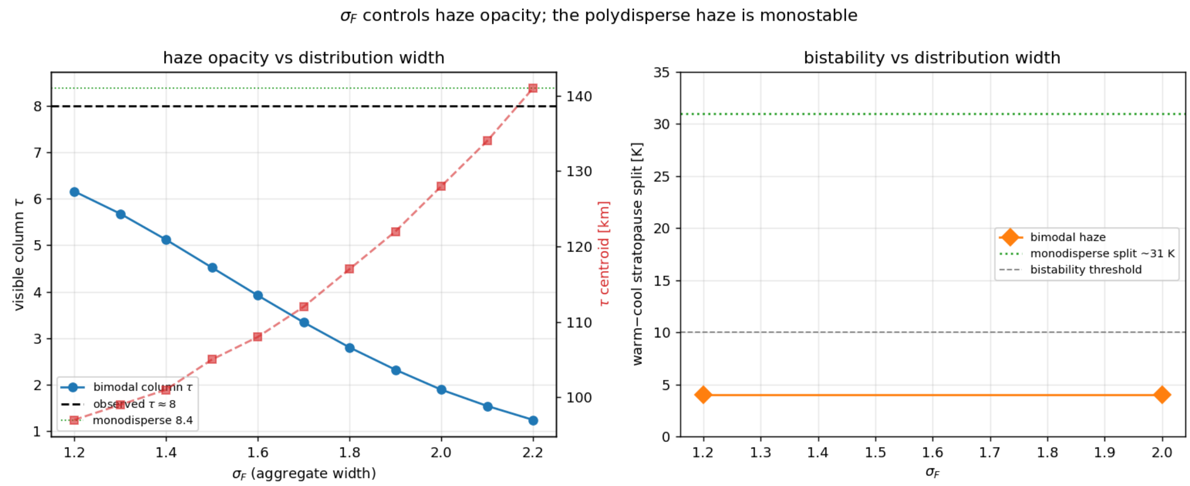

Figure 5: Bimodal (polydisperse) haze versus the aggregate geometric width σ_F. Left: the visible column optical depth falls steeply with σ_F (gravitational sorting), reaching the observed $\tau \approx 8$ only for a narrow distribution $\sigma_F \lesssim 1.2$; the τ-weighted centroid (dashed, right axis) rises as the haze thins. Right: in the lab-tholin configuration of §5.2, with the haze taken at the nominal temperature, the warm−cool stratopause split (the multiplicity indicator) stays near 3–4 K across σ_F, far below the monodisperse 17 K: a realistic size distribution strongly suppresses the frozen-haze multiplicity of that regime.

5.1 Implementation and the coupled iteration

We drive the Step 1 engine with the Step 2 haze through its prescribed-haze input: each pass maps the microphysics to the engine's per-band, per-level optical-depth tables (the same mean-field RDG cross-section used in §3, with the single-scattering albedo and asymmetry kept at the observational spectral shape; the longwave is a pure absorber whose opacity model §5.4 examines), re-runs the model to radiative–convective equilibrium, and feeds the new $T(z)$ back to the microphysics, under-relaxed. The interface is validated end-to-end: writing the *observational* haze through the same path reproduces the stock prescribed $T(z)$ to 0.000 K, and the microphysics-haze tables reproduce the optical depth of the in-line DISORT coupled run to machine precision. The engine top is first stabilised (a per-layer time-step cap reduces its $\lesssim$4 Pa oscillation from $\sim$24 to $\sim$2.5 K), since the haze source sits there.

5.2 First result, with laboratory-tholin IR opacity: a spurious strong-feedback regime

Our first coupled runs computed the haze longwave opacity from the RDG monomer absorption with laboratory tholin optical constants (Khare et al., 1984, hereafter the *lab-tholin* configuration). In that configuration the coupled iteration does *not* settle: under naive under-relaxation it oscillates between a *warm* branch with the stratopause near 183 K at 8 Pa and a *cool* branch near 131–135 K at 1.3–1.7 Pa, with a sharp transition—near $T \approx 137$ K a $\sim$1 K change in the input profile flips the equilibrium output by $\sim$48 K—the signature of a feedback with gain > 1. Holding the haze *fixed* at its transition-state value, the radiative–convective relaxation is genuinely multi-valued: warm and cool initial profiles land on distinct equilibria, 158 versus 141 K at the stratopause (17 K; up to 31 K deeper), reproduced in two independent RT engines; a bimodal polydisperse haze halves the frozen-haze split ($\sim$8 K, §5.3). A damped branch-tracking solve of the full coupled map $T = F(\mathrm{haze}(T))$ (the method of §5.4) nevertheless found a *single* coupled fixed point, stratopause $\sim$ 142 K, for every closure: the haze response removed the warm branch, and the loop's oscillation was overshoot of a steep ($|\mathrm{slope}| \gg 1$) but single-valued composite map.

Two features of this regime co-exist uneasily: a near-bistable feedback, and a coupled state 50–60 K *colder* than the prescribed-haze references (195 K in the Fortran engine, ~ 200 K in the same DISORT engine that hosts the coupled solve). As we show next, both have a single cause—the haze IR opacity—and neither survives its observational calibration.

5.3 Anatomy of the lab-tholin regime

Within the lab-tholin configuration, the composite map is $T \to \text{haze} \to T$; isolating each half locates the sharp transition (details and figures in SI S4). The microphysics half is *smooth*: sweeping input temperature profiles across stratopause 128–225 K, the haze column optical depth, τ-weighted centroid, and mass column all vary continuously (largest adjacent change $\leq 3.5\%$), the centroid rising smoothly with temperature—the positive feedback. The radiative half is genuinely *multi-valued*: holding the haze fixed at its transition-state value, warm and cool initial profiles land on distinct radiative–convective equilibria (158 versus 141 K at the stratopause, up to 31 K deeper), stable as the residual falls and reproduced in two independent engines—the absorbing-aerosol multiple-equilibria mechanism, driven self-consistently by the microphysics. Polydispersity already weakens it: with the bimodal haze of §4.7 the frozen-haze split halves (to $\sim$8 K; 3–4 K at the nominal haze, Fig. 5, right), the over-concentrated opacity of the single-size closure being part of the feedback's strength. The decisive question is whether the material properties that place the model in this dramatic regime are Titan's; the next section shows they are not.

5.4 The haze IR opacity is the decisive material property

The lab-tholin coupled fixed point sits ~ 40 K below the prescribed-haze equilibrium of the same RT. Comparing the two hazes source-by-source locates the cause on the longwave side (Fig. 6): the model haze actually carries *more* visible opacity than the observational haze above the stratopause (2.6× at 100 Pa), but its RDG infrared absorption—Khare laboratory-tholin k values—exceeds the observational haze LW opacity by a factor 20–90 across the 300–900 cm^{-1} window where a $\sim$150 K stratosphere emits, and is nearly gray where the real haze is strongly spectral. Per unit shortwave heating, the model stratosphere was cooling an order of magnitude too efficiently ($T \propto (a_{\rm SW}/\varepsilon_{\rm LW})^{1/4}$): the entire cold bias, and with it the strong-feedback regime, traces to one material property.

We therefore calibrate the haze LW opacity to the observations, in the same way the shortwave already takes $\omega_0(\lambda)$ and $g(\lambda)$ from the observational haze: the spectral absorptivity per unit visible extinction, $\chi(\tilde{\nu}, P) = \mathrm{d}\tau_{\rm LW}/\mathrm{d}\tau_{\rm vis}$, is read from the observational tables (it is nearly P-independent below 300 Pa—a composition property) and applied to the model haze's visible-extinction profile. The single change moves the monodisperse radiative–convective equilibrium from 141.5 to 194.2 K (+53 K), with the stratopause at 8.5 Pa, the same level as the prescribed-haze state.

With the calibrated haze, the branch-tracking (continuation) solve of the coupled fixed point $T = F(\text{haze}(T))$—a damped iteration $T_{k+1} = (1-\alpha)T_k + \alpha F(\text{haze}(T_k))$ with the haze recomputed from the current iterate every pass and the inner relaxation initialized from the current iterate, so a chain tracks the branch continuously connected to its start—is re-run from warm and cool starts for all three closures (Fig. 7). The coupled system is again *monostable*, but the regime is transformed. The composite map, steeply decreasing ($|\text{slope}| \gg 1$) in the lab-tholin configuration, is now nearly *flat* (slope ≈ 0.1): F returns 193–196 K whether the input stratopause is 163 or 200 K, for every closure, and undamped iteration ($\alpha = 1$) converges in a few passes. All six chains settle at a stratopause of ~ 194 K (monodisperse warm/cool: 193.7/193.6 K, a 0.1 K split; the bimodal chains agree within ±2–3 K, limited by an intermittent relaxation instability near the 1 Pa top, not by a second attractor). Re-running the frozen-haze

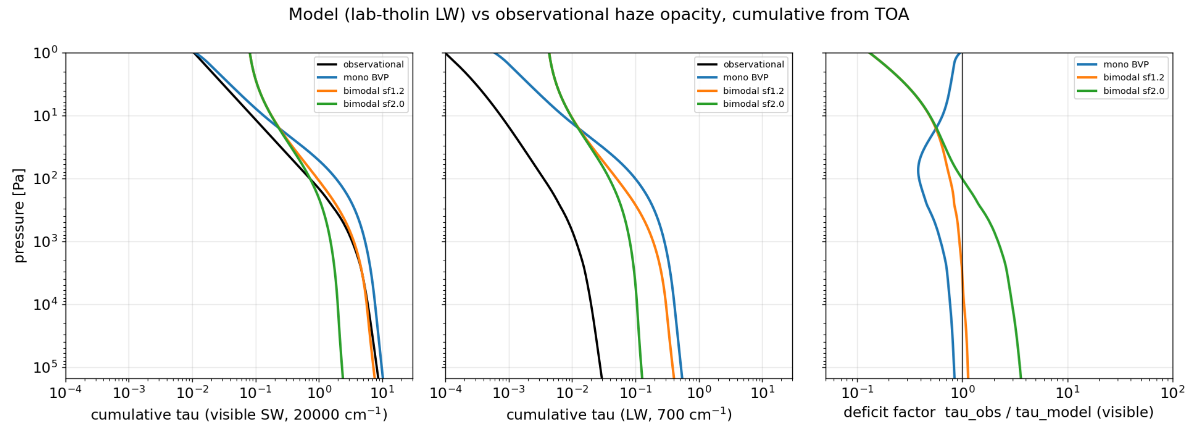

Figure 6: Why the lab-tholin coupled state is 40 K cold: cumulative haze optical depth from the top of atmosphere, observational (black) versus model hazes. Left: in the visible the model haze matches or exceeds the observational opacity at every level. Centre: in the thermal IR (700 cm^{-1} shown) the RDG/lab-tholin LW opacity exceeds the observational haze by 20–90×—the stratosphere cools an order of magnitude too efficiently per unit shortwave heating. Right: the visible deficit factor $\tau_{\mathrm{obs}}/\tau_{\mathrm{model}}$; the residual 2.6× monodisperse excess aloft is the leading remaining microphysics–observation difference (largely removed by the bimodal closure, orange).

warm/cool test of §5.3 with the calibrated haze likewise finds no robust multiplicity (stratopause splits ≤ 6 K, of either sign, for both closures): with realistic IR opacity the haze–climate coupling on Titan is *benign*—a single, strongly attracting state, no oscillation, no multiplicity to suppress.

The coupled fixed point lands in agreement with the prescribed-haze references themselves (195 K Fortran, ~ 200 K DISORT; 1 K below the former, 6 K below its same-engine DISORT reference): the residual microphysics–observation haze differences (the 2.6× visible excess aloft, Fig. 6, right) now move the stratopause by less than the inter-engine spread of the reference itself. The summary lesson is a sharpened version of the polydispersity caution of §4.7: the *feedback regime* of a haze–climate model is set by the haze IR/visible opacity ratio. Laboratory-tholin infrared optical constants place the model in a spurious strong-feedback regime—oscillating coupled iterations, fixed-haze multiple equilibria, a 40–50 K cold bias—all of which vanish when the haze IR opacity is calibrated to what Titan's stratosphere actually requires.

6 Outlook

Three refinements follow directly from the converged state. *(i) The residual haze-profile differences.* The coupled fixed point sits within the inter-engine spread of the prescribed-haze reference; the leading remaining microphysics–observation difference is the monodisperse visible-opacity excess aloft, which the bimodal closure already reduces to 0.96–1.3× of the observational profile below 100 Pa (§4.7), leaving mainly the 1–10 Pa production zone. *(ii) Charge inhibition.* The Coulomb coagulation barrier ($n_e = 15\, e^- \mu\mathrm{m}^{-1}$; de Batz de Trenquelléon et al., 2025) is implemented and measured: it freezes aggregate growth beyond $r_a \sim 0.3\,\mu$m (bringing the mean monomer count at 100 Pa from 3×10^4 down through the observed $\sim 3 \times 10^3$), but with the production rate held fixed the slower-settling haze accumulates to 5–13× the observed column while moving the stratopause < 1.5 K. Charge therefore enters only *co-tuned* with production and sticking efficiency—a recalibration of the Step-2 baseline, not a stratopause lever. *(iii) Photochemistry (Step 4).* In Steps 1–3 the production rate $\mathcal{Q}_p$ and seed properties are imposed ($\mathcal{Q}_p = 2.1 \times 10^{-13}\,\mathrm{kg\,m^{-2}\,s^{-1}}$ at $z_0 \approx 415$ km; de Batz de Trenquelléon et al., 2025). Step 4 promotes $\mathcal{Q}_p$ to an output of a photochemical module driven by CH_4/N_2

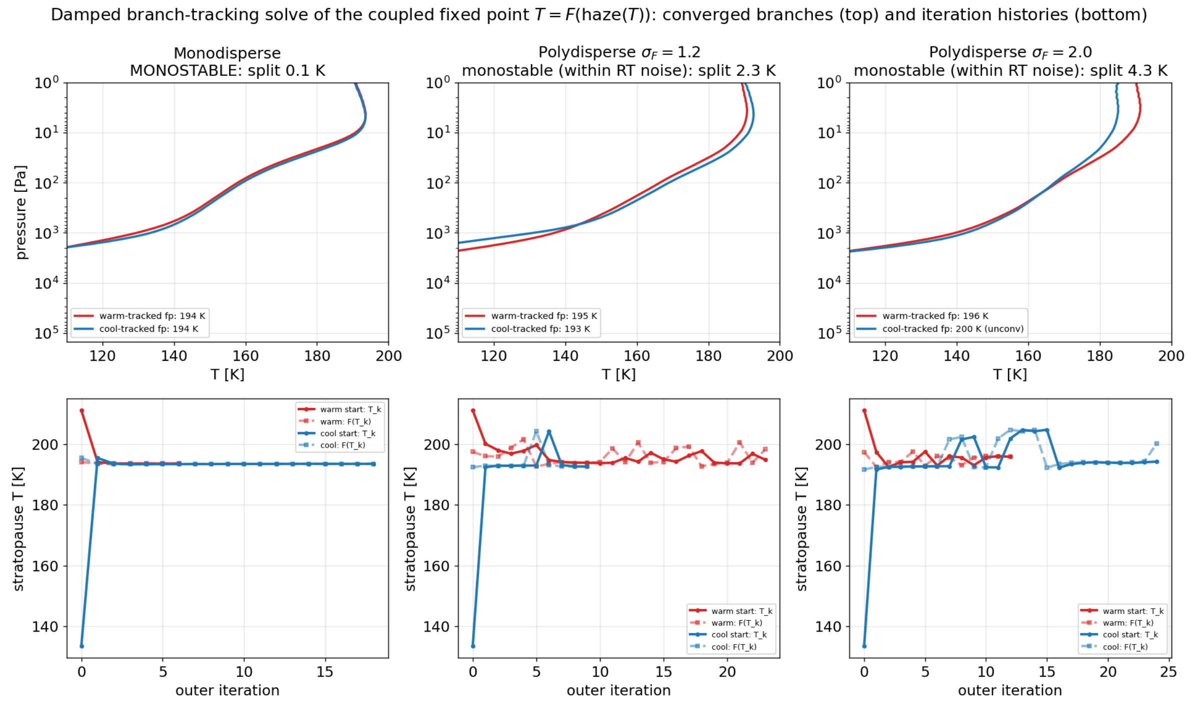

Figure 7: Branch-tracking (continuation) solve of the coupled fixed point $T = F(\text{haze}(T))$ with the observationally calibrated haze LW opacity, for the monodisperse haze (left) and the bimodal haze at $\sigma_F = 1.2$ (centre) and 2.0 (right). Top: warm-tracked (red) and cool-tracked (blue) chains converge to the same ~ 194 K stratopause for every closure (monodisperse split 0.1 K; bimodal agreement ± 2–3 K, set by an intermittent inner-relaxation instability near the 1 Pa top). Bottom: stratopause histories (circles: the iterate T_k; squares: its image $F(T_k)$); the image is 193–196 K regardless of the input—the composite map is nearly flat (slope ≈ 0.1), in sharp contrast to the steep lab-tholin map.

photolysis, so the haze mass flux responds to the radiation field and composition, completing the radiation↔chemistry↔microphysics feedback—now on a benign, easily converged coupled system.

Approximations. The monodisperse closure replaces the true size spread by $\bar{N}$ and evaluates the kernel at the mean size (the bimodal extension of §4.7 relaxes this); the spherical→fractal switch at $\bar{N} = 1$ is treated as instantaneous. Background profiles $\eta(T)$, $\lambda(T, P)$ are taken from N_2 kinetic theory, $g(z)$ from Titan gravity, and the eddy diffusivity $K(z)$ from a standard Titan profile (Vuitton et al., 2019). A full population balance is the main physics refinement beyond these. Baseline parameters are collected in SI S5.

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Supplementary Information

S1 Radiative-transfer implementation details

Longwave. Two opacity sources are combined on a common 40-band IR grid (0–2000 $\mathrm{cm^{-1}}$, 50 $\mathrm{cm^{-1}}$ bands). First, collision-induced absorption—the dominant thermal-IR continuum on Titan—from the HITRAN CIA coefficients (Karman et al., 2019) for N_2–N_2, N_2–CH_4, CH_4–CH_4, and N_2–H_2, as $\tau(\nu) = \sum_{\mathrm{pairs}} k_{AB}(\nu, T)\, n_A n_B\, \Delta z$ (band-averaged), or via the reference model's exponential-sum transmission fits when matching it (SI S2); the N_2–N_2 roto-translational band makes the far-IR strongly opaque (column $\tau \gtrsim 100$ near 50–100 $\mathrm{cm^{-1}}$). Second, correlated-k gas lines for the stratospheric coolants CH_4, C_2H_2, C_2H_6, C_2H_4, and HCN (10-point Gauss grid), summed under the perfectly-correlated assumption with each gas's VMR profile. Gases, CIA, and the pure-absorber IR haze are solved per (band, Gauss-point) with a band-integrated Planck source.

Shortwave. Methane opacity uses the same visible CH_4 k-table as the reference Fortran model (42 bands × 10 Gauss points, on a 20×10 pressure–temperature grid). Per layer the gas optical depth is $\tau = k(P, T)\, u$ with u the CH_4 column abundance in km-amagat; all 420 (band, Gauss-point) sub-intervals are solved in a single multi-wave DISORT call and recombined by the Gauss weights, with the per-band solar flux integrated from the Houghton spectrum and scaled to Titan's orbit (9.5 AU, $\sum$ bands $= 15.1\,\mathrm{W\,m^{-2}}$; diurnally-averaged $\cos\theta = 1/\pi$). The CH_4 windows let sunlight through to the surface while the bands absorb aloft; CH_4 alone absorbs $\approx 1.4\,\mathrm{W\,m^{-2}}$, peaking at ~ 14 K per Titan day in the stratosphere.

Relaxation to equilibrium. The *layer-mean* temperature is relaxed toward radiative–convective equilibrium: each iteration runs both bands, nudges the layer temperatures along the net heating rate (under-relaxed; the layer-mean state is in 1:1 correspondence with the per-layer heating, avoiding the level↔layer aliasing that otherwise seeds a grid-scale zig-zag), then applies a convective adjustment to a critical lapse rate ($\sim 1\,\mathrm{K\,km^{-1}}$). Cold space is imposed as the top boundary (a thermal emission temperature of 2.7 K, not the atmosphere's top temperature); without this the column spuriously absorbs downwelling longwave and overheats. The two layers nearest the 1 Pa model top retain a ±few-K $\mathrm{day^{-1}}$ residual zig-zag (a known boundary artifact, excluded from convergence metrics).

S2 Cross-comparison with the reference model: per-source opacity breakdown

To localise the residual heating-rate differences between the DISORT implementation and the reference Fortran model we compare the two engines' optical depth *source by source* at the same state (the reference's converged $P(T)$ profile and the shared prescribed haze), accumulating each contribution—haze, CH_4 shortwave gas, IR gas lines (CH_4/C_2H_2/C_2H_6/C_2H_4/HCN), Rayleigh, and CIA—separately from the top down (Fig. 8). For this we enabled the reference model's per-band Gauss-weighted gas-opacity output, so even the correlated-k gas can be compared directly rather than inferred.

The prescribed haze, which both models read from the same observational tables, interpolates identically: at matched band centres the per-band optical depth agrees to machine precision (ratio 1.000) in both the shortwave and the longwave, at every level. The CH_4 correlated-k gas likewise agrees at the column scale (shortwave 1.03, longwave 1.08). The one source that genuinely differed is the collision-induced absorption (CIA), which lives in the deep atmosphere ($P \gtrsim 10$ kPa): there the breakdown showed our continuum running well above the reference's.

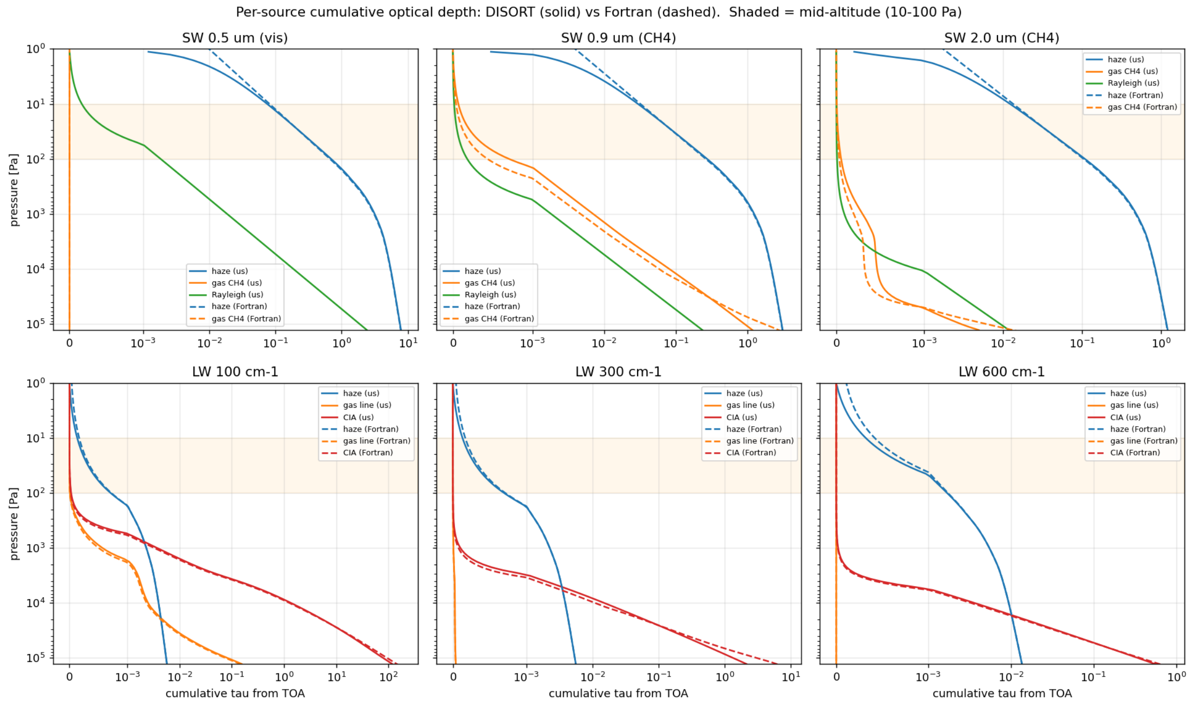

Figure 8: Cumulative optical depth from the top of atmosphere, by source, for our DISORT optics (solid) and the reference Fortran model (dashed), at the same state. Top row: shortwave (haze, CH_4 gas, Rayleigh) in three representative bands; bottom row: longwave (haze, gas lines, CIA). The shaded band marks the mid-altitude 10–100 Pa region. The haze dominates the mid-atmosphere and matches closely; the CIA continuum (bottom row, deep atmosphere) matches once our model adopts the reference's exponential-sum transmission fits in place of band-averaged HITRAN cross-sections.

We traced this to the CIA formulation. Our model band-averaged the HITRAN cross-sections and formed $\tau = \sum_{\text{pairs}} k\, n_A n_B\, \Delta z$, whereas the reference evaluates a three-term exponential-sum fit to the band *transmission*, $\mathcal{T} = \sum_i a_i \exp(-k_i u)$ with $\tau = -\ln \mathcal{T}$ and absorber amount $u = n_1 n_2\, \Delta z$—a form that captures the sub-band absorber-amount nonlinearity a single band-averaged cross-section cannot. On the shared Titan far-IR continuum the band-averaged cross-section overestimated the optical depth by $\sim 1.75\times$. Adopting the reference's exponential-sum fits (reading the same transmission tables) brings the two continua into agreement to a few percent through the bulk of the deep atmosphere (Fig. 8, bottom row; the per-band cumulative-τ ratio is 1.00 at 30 kPa, with the residual confined to the single lowest layer). With this, *every* opacity source—haze, CH_4, gas lines, Rayleigh, and CIA—matches the reference to within a few percent through the column, so the engines now differ only in the radiative-transfer solution itself, not in the optical depths fed to it.

S3 Microphysics: governing boundary-value problem and solvers

Working with the aggregate number density n and the monomer-volume density $M \equiv n\bar{N}$ (mass $= m_1 M$, $m_1 = \rho_p \frac{4}{3}\pi d^3$), the steady 1-D balance for any density q with downward settling speed ω and eddy diffusivity $K(z)$ reads $\mathrm{d}_z[\omega q + K\, \mathrm{d}_z q] = L_q - S_q$, where $\omega q + K\, \mathrm{d}_z q$ is the downward flux, S_q the production, and L_q the loss. Applied to the two densities,

$$\frac{\mathrm{d}}{\mathrm{d}z}\left[\omega(\bar{N},z)\,M + K\,\frac{\mathrm{d}M}{\mathrm{d}z}\right] = -\,S_M(z), \tag{10}$$

$$\frac{\mathrm{d}}{\mathrm{d}z}\left[\omega(\bar{N},z)\,n + K\,\frac{\mathrm{d}n}{\mathrm{d}z}\right] = \tfrac{1}{2}\,\beta(\bar{N},z)\,n^2 - S_n(z). \tag{11}$$

Equation (10) expresses monomer conservation: coagulation merges particles but preserves total monomer volume, so the only monomer source is production. Equation (11) expresses Smoluchowski number loss—each collision removes one aggregate at rate $\frac{1}{2}\beta n^2$—plus seeding. With $\bar{N} = M/n$ and $D = D(\bar{N})$, this is a coupled two-point boundary-value problem for $M(z), n(z)$. The production source is a Gaussian (de Batz de Trenquelléon et al., 2025, Eq. 5),

$$S_M(z) = \frac{\mathcal{Q}_p}{m_1}\,G(z), \quad G(z) = \frac{1}{\sqrt{2\pi}\,\Delta z}\exp\left[-\frac{(z-z_0)^2}{2\,\Delta z^2}\right], \quad S_n = \frac{S_M}{N_{\mathrm{seed}}}, \tag{12}$$

with $N_{\mathrm{seed}} = (r_p/d)^3$. Boundary conditions are vanishing densities at the model top and settling-only deposition ($\mathrm{d}_z M = \mathrm{d}_z n = 0$) at the surface. A first integral of (10) well below the source recovers the constant downward monomer flux $\omega M + K\,\mathrm{d}_z M = P$, with $P = \mathcal{Q}_p/m_1$. The monodisperse system is solved by collocation (`solve_bvp`) with the $K \to 0$ master-ODE profile as the initial guess.

Bimodal (polydisperse) transport. The bimodal two-moment state $(M_0^S, M_3^S, M_0^F, M_3^F)$ obeys the same balance per moment, with moment-specific settling velocities $\langle\omega\rangle_k$ (gravitational sorting) and the quadrature coagulation tendencies as local sources. This stiff four-field system defeats collocation. A *pseudo-time relaxation* (finite-volume semi-discretization: upwind settling, centred eddy diffusion, closed top, settling-only surface deposition; marched with a banded stiff integrator from the $K \to 0$ profile) is implemented but remains impractically stiff at production resolution; the identified path is operator splitting (implicit tridiagonal advection–diffusion, sub-cycled local coagulation). Bimodal results in this work therefore use the $K \to 0$ profiles; the eddy correction they omit is bounded by the monodisperse case, where restoring $K(z)$ moves the extinction scale height from 54 to 64 km.

S4 Anatomy of the lab-tholin strong-feedback regime

Supporting detail for §5.3. *Microphysics smoothness*: sweeping a family of input temperature profiles (stratopause 128–225 K) and recomputing the haze for each, the column optical depth, the τ-weighted centroid pressure, and the haze mass column vary continuously—largest adjacent change $\leq 3.5\%$, with no feature at the coupled transition near 137 K (Fig. 9). The centroid rises smoothly with temperature (1714 → 770 Pa across the sweep): a warmer stratosphere lifts the absorbing haze, which absorbs higher and warms further. *Radiative multiplicity*: at the frozen transition-state haze the warm/cool-start equilibria separate by 17 K at the stratopause (158 vs 141 K; up to 31 K deeper in the profile), the separation stable as the heating residual falls to $0.3\,\mathrm{K\,day^{-1}}$ and reproduced in both the Fortran and DISORT engines. Near the transition the coupled map is steep: a ~ 1 K input change flips the output by ~ 48 K. A caveat found while converging the branch-tracking chains: the radiatively sluggish lower stratosphere ($\sim$ 50–200 km) retains 10–18 K of initial-condition memory at the standard residual tolerance; a tightened-tolerance relaxation drifts it monotonically toward the common state with no barrier—convergence memory, not a second attractor.

S5 Baseline parameters

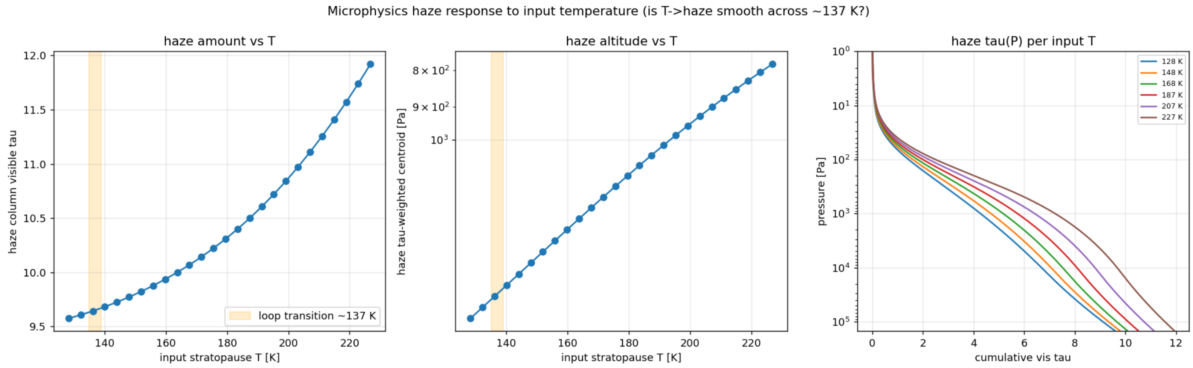

Figure 9: Microphysics haze response to the input temperature (lab-tholin configuration). Haze column optical depth (left), τ-weighted centroid pressure (centre), and cumulative $\tau(p)$ (right) all vary smoothly across the coupled transition (shaded, $\sim 137\,\mathrm{K}$): the $T \to$haze map has no cliff, so the multiplicity is radiative, not microphysical.

Table 2: Baseline microphysical and atmospheric parameters adopted for Titan. Sources: T08 (Tomasko et al., 2008), L23 (Lombardo and Lora, 2023), BR17 (Burgalat and Rannou, 2017), dT25 (de Batz de Trenquelléon et al., 2025).

Quantity	Symbol	Value	Source
Fractal dimension	D	2.0 (free, 2–3; 3 $\Rightarrow$ spheres)	dT25
Monomer radius	d	$50\,\mathrm{nm}$	dT25 (T08)
Production seed radius	r_p	$10\,\mathrm{nm}$	dT25
Aerosol density	ρ_p	$800\,\mathrm{kg\,m^{-3}}$	dT25
Mass production rate	$\mathcal{Q}_p$	$2.1 \times 10^{-13}\,\mathrm{kg\,m^{-2}\,s^{-1}}$	dT25 (T08)
Production altitude	z_0	$415\,\mathrm{km} \approx 1\,\mathrm{Pa}$	dT25
Production width	Δz	$20\,\mathrm{km}$	dT25
Charge density	n_e	$15\,e^-\,\mu\mathrm{m}^{-1}$	dT25
Slip-flow constant	A	1.591	BR17
Surface temperature	T_s	$93.5\,\mathrm{K}$	T08
Surface pressure	P_s	$\sim 1.47\,\mathrm{bar}$	L23
Stratospheric CH_4	–	1.4%	T08
Haze opacity scale height	H	$65\,\mathrm{km}$ (above $80\,\mathrm{km}$)	T08
RT grid	–	0–$540\,\mathrm{km}$, ~ 50 layers	T08, L23
Shortwave / longwave split	–	$5\,\mu\mathrm{m}$	L23